

BEAUFORT MCAS, SC, USA

WMO: 722085

Lat: 32.483N

Lon: 80.717W

Elev: 37

StdP: 14.68

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 93831

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 28.0	(c) 31.7	(d) 10.5	(e) 9.4	(f) 36.3	(g) 15.6	(h) 12.2	(i) 39.3	(j) 22.2	(k) 54.6	(l) 19.8	(m) 52.0	(n) 5.2	(o) 350	(p) 0.390

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 16.1	(c) 95.0	(d) 78.3	(e) 92.8	(f) 77.9	(g) 90.9	(h) 77.6	(i) 81.5	(j) 88.6	(k) 80.5	(l) 87.7	(m) 79.6	(n) 87.0	(o) 7.1	(p) 270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
79.5	153.1	84.7	78.8	149.5	84.2	77.5	143.2	83.3	45.0	88.8	44.0	88.2	43.0	86.8	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 18.6	(b) 16.5	(c) 14.5	(e) 22.9	(f) 99.2	(g) 3.2	(h) 2.7	(i) 20.6	(j) 101.1	(k) 18.8	(l) 102.7	(m) 17.0	(n) 104.2	(o) 14.7	(p) 106.2		
(a) 18.6	(b) 16.5	(c) 14.5	(e) 20.0	(f) 83.8	(g) 3.0	(h) 1.6	(i) 17.9	(j) 85.0	(k) 16.1	(l) 85.9	(m) 14.4	(n) 86.8	(o) 12.3	(p) 87.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	67.3	49.9	53.4	59.2	66.6	74.2	80.2	82.8	81.8	77.5	68.8	58.7	53.2
	DBStd	13.39	9.28	8.43	8.09	6.64	5.52	4.10	2.98	3.13	4.55	6.87	8.01	8.97
	HDD50	279	119	55	19	0	0	0	0	0	0	0	17	69
	HDD65	1722	472	334	211	60	5	0	0	0	1	43	219	377
	CDD50	6582	115	149	304	500	752	906	1016	986	826	581	278	169
	CDD65	2551	4	8	32	109	292	456	551	521	377	160	30	11
	CDH74	23303	11	44	200	703	2333	4448	5956	5341	3096	1013	127	31
	CDH80	8661	0	2	27	142	751	1781	2567	2147	1028	206	10	0
Wind	WSAvg	6.4	6.7	6.9	7.7	7.5	6.8	5.9	5.7	5.3	5.9	5.9	6.0	6.2
Precipitation	PrecAvg	50.6	3.5	3.3	3.7	3.0	3.5	6.0	6.4	7.3	4.5	2.5	2.3	3.1
	PrecMax	67.6	8.8	7.8	8.9	7.1	10.5	14.6	19.1	18.0	12.3	9.6	7.8	6.3
	PrecMin	33.4	0.2	0.2	0.6	0.1	0.2	0.9	1.5	1.3	0.6	0.1	0.0	0.1
	PrecStd	9.1	2.0	1.8	2.2	1.8	2.3	3.3	3.6	3.9	3.1	2.3	1.9	1.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	75.3	79.1	83.4	88.2	94.2	97.3	97.9	97.2	93.0	88.2	81.6	77.7
		MCWB	65.5	66.6	66.0	68.8	74.5	77.5	78.2	80.0	76.8	75.6	71.9	68.8
	2%	DB	71.6	74.7	79.2	83.7	90.1	93.9	94.9	93.5	90.2	84.3	77.7	73.8
		MCWB	63.6	64.0	65.2	69.3	73.4	77.3	78.7	78.8	76.7	73.5	68.4	66.7
	5%	DB	67.6	70.6	75.4	80.9	87.1	91.2	92.8	91.3	88.1	82.1	74.6	70.2
		MCWB	61.2	61.4	63.8	68.0	72.9	77.1	78.7	78.7	76.0	72.2	66.6	65.3
	10%	DB	64.2	66.8	72.2	78.3	84.1	88.7	90.6	89.5	85.9	79.9	72.0	67.2
		MCWB	59.0	59.5	62.3	66.9	71.9	76.4	78.5	78.2	75.0	70.8	65.8	63.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	68.5	69.5	70.4	74.8	78.3	82.1	82.9	83.0	81.0	79.4	74.1	71.0
		MCDB	72.2	75.1	77.9	80.5	87.2	90.9	90.9	90.4	85.9	82.8	77.9	74.4
	2%	WB	65.5	66.5	68.4	72.6	76.6	80.1	81.4	81.5	79.4	77.0	71.4	68.5
		MCDB	69.4	71.9	75.0	79.3	84.4	88.8	89.0	88.2	84.9	81.7	74.7	71.7
	5%	WB	63.2	64.0	66.4	70.8	75.4	78.9	80.4	80.5	78.2	75.2	69.3	66.2
		MCDB	66.6	68.2	72.1	77.5	82.7	87.2	88.3	87.2	84.1	79.9	73.0	69.7
	10%	WB	60.3	61.3	64.5	69.2	74.2	77.9	79.5	79.6	77.3	73.3	66.7	63.6
		MCDB	63.7	65.5	69.7	75.5	81.1	85.7	87.7	86.4	83.3	77.6	70.7	66.6
Mean Daily Temperature Range	5% DB	MDBR	19.3	19.8	20.0	19.8	18.4	16.6	16.1	15.3	15.3	18.0	19.9	18.7
		MCDBR	21.4	22.9	22.9	21.5	21.0	19.4	18.4	17.5	17.4	18.3	20.8	19.6
		MCWBR	14.8	14.2	12.0	10.4	9.0	8.0	7.5	7.1	6.9	9.0	12.9	13.9
	5% WB	MCDBR	18.9	19.8	18.8	18.0	17.8	17.4	16.8	16.2	14.8	14.5	16.5	17.9
		MCWBR	14.8	14.1	11.4	10.0	8.7	8.2	7.6	7.3	6.7	8.2	11.9	14.0
Clear-Sky Solar Irradiance	taub	0.331	0.340	0.370	0.403	0.457	0.497	0.527	0.509	0.431	0.397	0.354	0.338	
	taud	2.526	2.495	2.430	2.330	2.219	2.154	2.088	2.138	2.354	2.417	2.494	2.534	
	Ebn at Noon	283	291	288	282	266	254	246	249	266	267	271	273	
	Edh at Noon	27	30	35	40	45	49	51	48	37	33	28	26	
All-Sky Solar Radiation	RadAvg	929	1137	1485	1886	2017	2024	1966	1783	1517	1323	1033	823	
	RadStd	48	113	134	121	133	144	123	116	125	140	70	67	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	+1.15	N/A	N/A	+0.92	N/A	N/A	N/A	-158	+418	+279
(47) Regional (37 neighbors)	+1.08	N/A	N/A	N/A	N/A	N/A	N/A	-180	+374	+261

Nomenclature: See separate page